33kV OVERHEAD LINE FROM PARKSTON
POWER STATION POLE FIM FA35 RECLOSER
2000-F00-SLD-EL-10001

INTERTRIPS TO FIM FA35 RECLOSER

F12-NR-012
PRIMARY CRUSHING TRANSFORMER
F12-TR-012 NEUTRAL EARTHING CUBICLE

F12-TR-012
33 /11 kV
5000 kVA
Dyn11
z=7%
KNAN

254Ω +/-5%
30/0.05A 25A FOR 10s
SENSING
ER-15kV

TRANSFORMER PROTECTION DEVICES TERMINAL BOX

F12-SR-001
11kV PRIMARY CRUSHING SWITCHROOM

F12-SW-012
PRIMARY CRUSHING 11kV SWITCHBOARD

NOTE 4

TEST

300/1A 2.5VA Cl 0.2 — PQM SEL735
300/1A 2.5VA 5P20 — PR SE-330AU

EARTH FAULT INDICATION
TRANSFORMER INSTRUMENTATION ALARM & TRIP CONTACTS (NOTE 6)
UPSTREAM TRIPS

NOTE 2
CLOSE / TRIP 630A
50 51 50G 51G AFD 50N 51N — PR SEL 751
49 63 71

F12SR001-LCS
E-SHUT
NOTE 5

A01 2000-F12-SCM-EL-10101
A02 2000-F12-SCM-EL-10103
A03
11kV 1250A 3-E 25kA FOR 3s
A04 2000-F12-SCM-EL-10107
A05 2000-F12-SCM-EL-10109

NOTE 2
NOTE 3

2A 630A
VT
11kV/√3 / 0.11kV/√3 / 0.11kV/√3
15VA CLASS 0.2
20VA CLASS 3P

160E
CLOSE
50 50G 27 AFD 51 51G 81 — PR SEL 751

100/1A 2.5VA 5P20
50/1A 1.5VA 10P10
TEST

VSD CONTACTOR CLOSE PERMISSIVE
VSD HEALTHY
VSD FAULT

F12CG401-VS
PRIMARY CRUSHER 2
11/3.3kV 1000kW VSD

M 3~
600 kW
RTD
F12-CG-401
PRIMARY CRUSHER 2
2000-F12-SCM-EL-10105

200/1A 2.5VA 5P20
50/1A 1.5VA 10P10

ACTIVATION OF SWRM E-SHUT
DETECTION OF EARTH FAULT
DETECTION OF ARC FAULT

F12-TR-001
PRIMARY CRUSHING
11/0.433kV TRANSFORMER

F12-SW-001
PRIMARY CRUSHING
415V MCC INCOMER

REFER DRG 2000-F12-SLD-EL-10201

11kV SPARE
FULLY EQUIPPED
-

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

Signed [redacted] Date 04/09/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

NOTES

1. THE SWITCHBOARD SHALL BE EXTENSIBLE ONE END FOR FUTURE EXPANSION AND SUPPLIED WITH ONE SPARE TIER AS SHOWN.
2. ARC FLASH IN INCOMER CIRCUIT BREAKER COMPARTMENT IS NOT CURRENT SUPERVISED.
   TRIPPING OF BUS AND DOWNSTREAM CB/FC COMPARTMENTS IS CURRENT SUPERVISED.
3. ARC FLASH POINT DETECTORS SHALL BE PLACED IN THE CABLE ZONE, CIRCUIT BREAKER CHAMBER OR FUSE CONTACTOR CHAMBER AND THE BUSBAR CHAMBER. THE DETECTORS SHALL BE CONNECTED DIRECTLY TO THE PROTECTION RELAY IN EACH PANEL.
4. ONE OFF 30/0.05A PROTECTION CT SHALL BE INSTALLED IN THE NER CUBICLE AND CONNECTED TO THE INCOMER PROTECTION RELAY TO MONITOR FOR EARTH FAULT DETECTION .
5. SWITCHROOM EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON IS HARDWIRED DIRECT TO THE SHUNT TRIP OF THE INCOMER CIRCUIT BREAKER AND THE CORRESPONDING UPSTREAM POLE MOUNTED RECLOSER.
6. 33/11kV TRANSFORMER INSTRUMENTATION TRIP CONTACTS SHALL BE HARDWIRED TO THE 11kV SWITCHBOARD INCOMER PROTECTION RELAY WHICH SHALL INTERTRIP THE UPSTREAM 33kV FEEDER.
7. UPSTREAM 11kV FEEDER SHALL BE TRIPPED ON THE EARTH & ARC FAULT DETECTION OF THE DOWNSTREAM INCOMER ACB.

LEGEND

PR	INCOMER / FEEDER PROTECTION RELAY
PQM	POWER QUALITY METER
	ARC FLASH SENSOR (POINT TYPE) NOTE 2, 10
	CAPACITIVE VOLTAGE INDICATION
	MECHANICAL INTERLOCK DEVICE
E-SHUT	SWITCHROOM EMERGENCY SHUTDOWN
M	MOTOR OPERATOR
27	UNDER VOLTAGE
50	INSTANTANEOUS OVER-CURRENT
50G 50N	INSTANTANEOUS EARTH/NEUTRAL OVERCURRENT
51	TIMED OVER-CURRENT
51G 51N	TIMED EARTH/NEUTRAL OVERCURRENT
59	OVER VOLTAGE
64	EARTH FAULT
49	TRANSFORMER TEMPERATURE
63	TRANSFORMER PRESSURE / BUCCHOLZ
71	TRANSFORMER LIQUID / GAS LEVEL
81	FREQUENCY
AFD	ARC FLASH DETECTION

2000-F12-SLD-EL-10001_Rev1
Single line diagram

O:\CAD PROJECTS\2395 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\ELE\F12 - PRIMARY CRUSHING\2000-F12-SLD-EL-10001.DWG 13AUG25 11:50AM HOTSEAT04